

The Multivariate Normal Distribution

A random vector is normal when every linear combination of its entries is normal. From that definition come the moment generating function, the density, and closure under all linear maps, singular ones included. Two further properties make the normal distribution the working assumption of linear-model theory: zero covariance between jointly normal blocks means independence, and conditioning one block on another gives a linear regression with constant error variance. Partial correlation, the correlation that remains after linear adjustment, connects these facts to least squares.

What you need. [Chapter 1](#) (spectral theorem, square roots, generalized and partitioned inverses), [Chapter 2](#) (mean vectors and covariance matrices). Some examples refer forward to least squares (Chapter 5 and [Chapter 6](#)).

3.-1.1 Roadmap

- [Definition through linear combinations](#)
- [Moment generating function and density](#)
- [Linear transformations and marginals](#)
- [Independence and zero covariance](#)
- [Conditional distributions](#)
- [Partial correlation](#)
- [Summary and notes](#)